ripwire

The ripgrep of AI context.

Paddle out with a map.

See the rip before you’re in it.

A zero-runtime-dependency C++23 CLI that maps any codebase into a ranked, deterministic call graph for coding agents — and puts a tripwire on every claim it emits.

rip — the speed half

Structural answers in tens of milliseconds, from a call graph it built itself.

wire — the other half: honesty

Unprovable totals ship labelled as floors. A zero means none found — never none exists.

Every truncation is disclosed, and every guess says how many it chose from.

Apache-2.0 · single binary · hermetic build (proven with the network off) · 23 vendored tree-sitter grammars

// the problem

Your agent reads the whole library to answer one question

Blind grep, then whole-file reads. Most of what enters the context window is never used — it is paid for anyway, on every step of every task.

And bigger context is not better context: irrelevant text actively degrades the answer. The job is selection — a small, ranked, structural map beats a pile of open files.

0.108 s

median answer, warm index — round 4, all arms one machine one day, N = 60

−39.4%

token ceiling vs the un-routed baseline — LocBench cost ledger, N = 243

Speed caveat travels with the number: ripwire answers from a warm, pre-built index, and round 4 measured its index cost too — 0.31 s, tabulated beside the competitors' rather than omitted.

// one binary, the whole pipeline

Crawl → parse → resolve → rank → emit. Deterministic, end to end.

crawl →	**parse** →	**resolve** →	**rank** →	**emit**
the tree, no VCS needed	tree-sitter, 23 vendored grammars	references → call graph	Personalized PageRank	minified XML, one line

byte-identical	two runs, same bytes — a gate on every push, not a tendency; warm equals cold
zero runtime deps	CMake + a C++23 compiler; builds with the network off — vendored everything
the languages	Rust · C++ · ObjC/C++ · C · Metal · CUDA · Python · Go · Swift · TypeScript · JavaScript · Java · Ruby · PHP · Lua · Elixir · Dart · Bash · C# · JSON · TOML · YAML · Markdown — 23 vendored grammars; markdown headings are real symbols
agent-native	an MCP server and 179 long flags behind one `--help` that is always the authority

// every feature, one screen

Seven families of questions it answers

understand cold	*What is this repo, and what matters in it?*	`--for` `--tree` `--lego` `--exemplar` `--recall` `--pack-task` `--token-budget`
navigate	*Who calls this? Safe to change or delete? Which tests?*	`--callers` `--callees` `--uses` `--impact` `--path` `--connect` `--affected` `--situ` `--test-gate` `--from-trace` `--pattern` `--safe-delete`
detail ladder	*Show me more — but only where it pays.*	`--detail` `--pack-signatures` `--outline` `--expand` `--compress`
quality & risk	*Where is the risk, and did I just add some?*	`--quality-panel` `--quality-delta` `--dmm` `--readability` `--ensemble` `--context-ratio` `--nonlocal-state` `--field-affinity` `--hotspots` `--lint` `--clones`
self-diagnosis	*Is my setup actually working?*	`--doctor` `--skipped`
security	*Is this agent skill file safe to install?*	`--scan-skill` `--scan-skills`
knobs & modes	*Shape, format, cache, budget.*	`--json` `--format` `--mcp`

--help is generated from the binary's own flag table — 179 long flags; docs/COMMANDS.md carries an entry for every one of them, 161 with a recorded invocation and its output

`// reach for it at the moment, not the manual`

The reflexes: which verb fires when

Landing cold on a task

`--for="<task>"` · `--pack-task`

ranked, budgeted context in one call

Stack trace in hand

`--from-trace`

frames mapped to symbols, innermost body included

“Is it safe to change X?”

`--impact` · `--uses`

the whole blast radius, not just 1-hop callers

Just edited a symbol

`--edit-check`

did the contract change, who breaks — ~ms, warm

About to write a helper

`--exemplar` · `--grep`

imitate the house pattern; catch the duplicate early

Landing several branches

`--merge-scout`

pairwise conflict sites + a landing order

Before you call it done

`--quality-delta` · `--test-gate`

only what you made worse; exactly which tests to run

Handing the area on

`--handoff` · `--note-add`

the verified-vs-heuristic packet the next agent needs

`and --pr-context when you are reviewing someone else's diff — per-file blast radius, tests-to-run, hotspot flags`

// new since 2026-08-15 — each figure re-derived on the binary that ships

What 2026-08-15 to 2026-08-23 added

`--pattern` *search by code SHAPE, not by node kinds*	foo($X, ...) across 13 grammar objects / 11 languages. On this repo VERIFY($X) finds 101 hits over 625 eligible files, each with its enclosing symbol. A pattern no served grammar resolves REFUSES (exit 1) instead of reporting hits=0.
`--safe-delete` *“can I delete this?” in one call*	callers + transitive radius + every read/write/import site + how much of that radius a test reaches. risk= names what was FOUND — never a go/no-go verdict.
`--impact importers=` *the second, weaker reach*	the files that include/require a file defining SYM, with their own cap pair — NEVER summed into reaches=, because files and symbols are different units. IngestResult: reaches=0, importers=70.
`compact --for bundles` *bodies were 52.7% of every conceptual byte*	the subtoken+body route now ships the ranked map plus one-hop edge context, disclosed as bundle=compact bodies=0. 15-query class B: 184,857 B → 95,256 B, −48.5%, all 11 decisive markers still present. --auto-bodies is a permanent opt-out.
`corroborated callers` *shared= on --pack-task rows*	a neighbour reached by four of the bundle's anchors sorts above one reached by a single anchor — omitted at 1, which is what every 1-hop row satisfies anyway.
`per-rule --lint roll-up` *a capped view stops hiding whole rules*	on this repository 14 of the 31 rules that fired contribute ZERO shown rows — 368 findings whose only evidence is their count=. A rule no corpus language registers carries applicable=0, so its zero is inertness, not a measurement.

And two languages: PHP and Lua, each shipped with its floor stated rather than implied — PHP's run-time dispatch ($fn(), call_user_func, __call) names its callee at run time and is a declared floor; a Lua corpus reports no inheritance edges, because metatable inheritance has no syntax to read.

// new since 2026-08-23 — each figure measured, dated, and re-derivable

What 2026-08-23 to 2026-08-30 added

--slice-flow

cross-statement data-flow slicing (ARISE, arXiv:2605.03117)

The paper's own slicer — reaching-definition edges, a seed plus a direction, a bounded BFS that stops at function boundaries — implemented and MEASURED on a registered fix-commit protocol: flow rows lift function-level added-line recall 0.163 → 0.198 at 25% of --expand's whole-body bytes (whose recall is 1.0 by construction). 7 commits / 38 instances, cpp only — a thin corpus, reported as thin.

--slice, first numbers

the 2026-08-28 registered contract, finally measured

Per-variable line-recall 0.726, hit-all rate 0.632 — a FLOOR under a noisy relevance oracle: every inspected miss was the protocol's word-regex matching identifiers inside comments and string literals, occurrences the slicer correctly refuses to call variable uses. No inspected instance showed a real occurrence the slice dropped.

~1350× on --expand

the secret-redaction sweep was quadratic in LINE length

One selector in a 2.1 MB / 768-line minified yarn bundle NEVER completed — killed at 1,343.9 s of user CPU with an empty output file; 196.7 s is the only clean lower bound — and now answers in 0.46 s warm. The 20 KB single-line fixture: 23.02 s → 0.017 s. A pure memoization, so an output no-op: 120/120 invocations byte-identical; gated on behaviour, the timings a ledger row and never a red-CI threshold.

sources: docs/EVALS.md §--slice-flow (registered protocol + per-instance ledger) · bench/PROFILE.md 2026-08-30 — no number travels without its caveat

// new since 2026-08-30 — the window this deck did not cover until now

What 2026-08-30 to 2026-09-06 added

.gitignore, honoured by default *named for ripgrep, whose defining default this is*	The crawl walked ignored files; in a git work tree it now consults git's own rules and skips what the repository already declared uninteresting. This root, three checkouts under an ignored bench/external/: files= 8,674 → 1,522, cold 2.81 s → 0.52 s, warm 0.63 s → 0.10 s — and --no-ignore restores the previous walk exactly. Nothing drops silently: ignored_files= is exact, ignored_dirs= says the walk STOPPED there so those contents are UNKNOWN rather than zero, and both are ABSENT when the rules dropped nothing — so a tree with nothing ignored is byte-identical to what it produced before.
--html --color-by *the map, rendered — no server, no CDN*	One self-contained HTML file: a force-directed call graph you click to recentre. --color-by=lang|community|cx|churn|tested sets the INITIAL colour only — the page embeds all five and keeps a live selector, so switching lens costs no second run. One five-stop blue-to-orange scale rather than the usual green-to-red, so reading it never depends on telling red from green.
--eval-retrieval, re-sampled *the sampler was measuring the corpus, not the ranker*	For fourteen months the gold set was the first 150 doc-commented symbols in PATH order — and bench/ sorts before src/, so adding a docstring to a benchmark harness moved a published number without touching ranking code. One 60-symbol probe, one corpus, only its PATH varied: 0.105 and 0.356 MRR from spelling alone. Now exhaustive — population=3011 scored=3011 rule=exhaustive — and EVERY retrieval figure this project publishes was re-derived under it.
--plan-lanes *which lanes would collide, BEFORE a line is written*	Where --merge-scout says "these branches already conflict", this says "these lanes WOULD conflict if assigned this way" — no ref to resolve, no re-ingest. Each lane now also carries an advisory execution profile (model + reasoning effort) that labels its own evidence basis="structural-only" and versions its policy separately from the schema, so a recommendation cannot quietly claim more than the structure it read.

CHANGELOG.md [Unreleased] carries the ignore-default measurement and its ledger in bench/PROFILE.md · the sampler negative is docs/EVALS.md §7, published as a counterexample against our own published numbers

// 0.6.0 — everything since v0.5.0 (tagged 2026-09-07); each figure names its PR

0.6.0: faster where it hurt, clearer where it stops

a new language

Dart — the 23rd grammar, by @calvinchengx: 71,726 symbols from flutter/packages' 3,706 .dart files, none among its 289 degraded parses (#75, #106)

any language — Dart's landing caught six per-language arrays sized off the last enum by hand: an appended language dropped out of --skipped's census silently. Fixed for all.

Rip'n Fast

159.7 → 9.2 s — warm --grep on llvm-project: an inheritance cone rebuilt on every still-ambiguous receiver-typed call is now built once per type (#83)

−19.9% CPU — cold parse on llvm-project, 194.1 → 155.6 s: child walks that went quadratic now use a cursor, output byte-identical (#127, #130)

−27.6% CPU — warm --pack-task on go, 8.13 → 5.88 s: the tokenizer rebuilt on one header of NEON, AVX2 and scalar string kernels (#127)

honest where it counts

2,107 → 3 — false callers of memgraph's SafeString::move: a std::move call no longer binds to your own code (#134)

12/12 → 8/12 — commits --quality-delta gated, of 12 landed: true-positive share 2% → 12%, with 0 wrong rows (#127)

13/25 → 0 — harmful --help-task recommendations on the adversarial prose set; precision 0.797 → 1.000 (#127)

fewer tokens, nothing hidden

46,385 → 4,473 — tokens for --help: one line per flag now, and every disclosure is still one call away (#92)

6 → 50 — --handoff symbols per code file (12 per prose file): containment 16% → 54%, purely additive (#127)

pages, not cuts — --doc-drift, --flags, --flip and --situ disclose their cuts and page; --situ lists every tests-to-run row (#127)

upgrade note: prebuilt x86-64 Linux binaries now need an x86-64-v3 (AVX2-class) CPU, RHEL 10's own floor; arm64 uses NEON (#127)

// Rip'n Fast, at scale — llvm-project, 182,555 files, as the instrument

Three bugs found at llvm-project scale

O(C²) child walks

An indexed ts_node_child(n, i) loop re-walks the child list on every call, so wide, flat lists (comment runs, C/C++ include guards) go quadratic. #127 moved 24 walks to one cursor helper (18 arms red first at 11–125×); #130 moved 22 more (13–126×).

194.1 → 155.6 s

cold parse CPU, n = 1
−19.9% · wall −25%

the inheritance cone

Rebuilt on every still-ambiguous receiver-typed call: 86,667 cones for 2,984 distinct types, 143 s of a 154 s run. Each type's cone is now built once, and default maps are byte-identical on go and llvm-project.

159.7 → 9.2 s

warm --grep
default map 248 s → 10 s

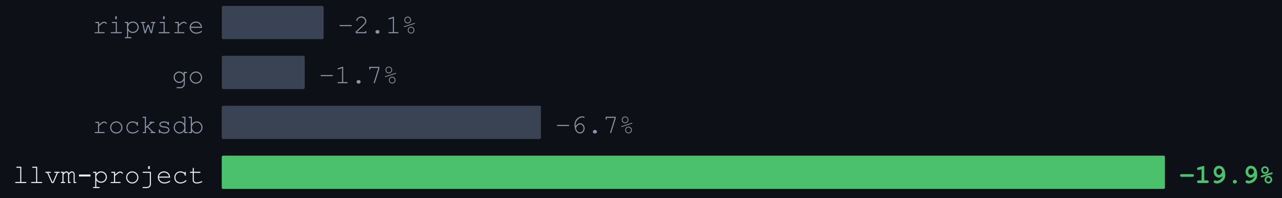
the cache that evicted itself

One llvm root needs 1.76 GB of cache, and the 2 GB oldest-first sweep was evicting that root's own index. The sweep now pins the root you are working in, and two key builders that hashed one root with different seeds share one key.

274 → 26 s

a repeated --for, CPU

the same round's cold parse, CPU Δ by corpus (median of 5; llvm-project n = 1)



Why llvm-project

A scale test that is also the real thing. Built on or with LLVM:

Clang (C, C++, Objective-C)
Chromium's only supported compiler; the Android NDK's; Apple Clang in Xcode; the PS4 toolchain

Swift · Rust · Julia · Zig · Kotlin/Native
each has an LLVM backend for its code generation

CUDA
NVIDIA's NVVM compiler is based on LLVM, and Clang compiles CUDA too

Metal
Apple's shading language: “Metal uses clang and LLVM”

MLIR
part of LLVM; used by TensorFlow, OpenAI Triton and Mojo

WebAssembly
Emscripten, a compiler toolchain to WebAssembly using LLVM

ROCm · oneAPI
AMD ROCm's compilers fork llvm-project; Intel's oneAPI DPC++/C++ uses LLVM

LLVM received the 2012 ACM Software System Award · each item's source is in the notes

all three measured on llvm-project; no standard corpus could see the first · sources: #127, #130, #83 — quotes in the notes

// wire, in practice — three stories from 0.6.0, all merged

Confidently wrong, caught, and fixed in the open

#134 · the resolver

std::move had 2,107 callers

On memgraph, calls written std::move bound to the repository's lone in-repo move, SafeString::move, at full confidence and with no amb= or prov= marker. It became map row #1. In ripwire's own graph, std::min, std::max and std::sort bound fastmath and svector members.

2,107 → 3

callers of SafeString::move after the fix: two of its unit tests and one std::ranges::move, a disclosed gap

what catches it now

A std::-qualified call keeps only candidates inside namespace std; anything else counts toward external= and gets no edge. On ripwire's own tree, --callers=min went from 131 to 5.

test/stdqualcheck.sh · 35 checks, 19 red before

#127 · the string kernels' gate

The probe that lied

strkerncheck asked Rosetta 2 whether it could run the x86-64-v3 kernels, using a one-add AVX2 probe. Clang folded that probe to a scalar addb, with zero ymm or VEX opcodes, so it answered ok on every runtime while the v3 slice SIGILLed on the macos-14 runner. The first fix's commit gave the wrong reason; the next one corrected it.

0 ymm opcodes

in the probe binary that had answered “ok” (otool)

what catches it now

The probe is now built with the floor's own -march and touches every v3 extension. The gate disassembles it and requires ymm, pdep, pext, lzcnt, tzcnt, vfmadd and vcvtph2ps; a folded probe routes to the scalar slice instead of passing.

test/strkerncheck.sh · b4ebf0bb, 150fb6d3

#127 · the audit's own tools

The audit's instrument was wrong twice

A 14× super-linear reading turned out to be the cache evicting its own working root. And the recorded cap sweep had measured retrieval caps on a population of zero: it discarded exit codes, counted rows that emit nothing, and read back its own 10.4 MB cache blob, which “responded” to 103 of 108 caps.

59/195 → 64/151

cap-sensitive rows: the published split, then the split re-derived over the rows that answer

what catches it now

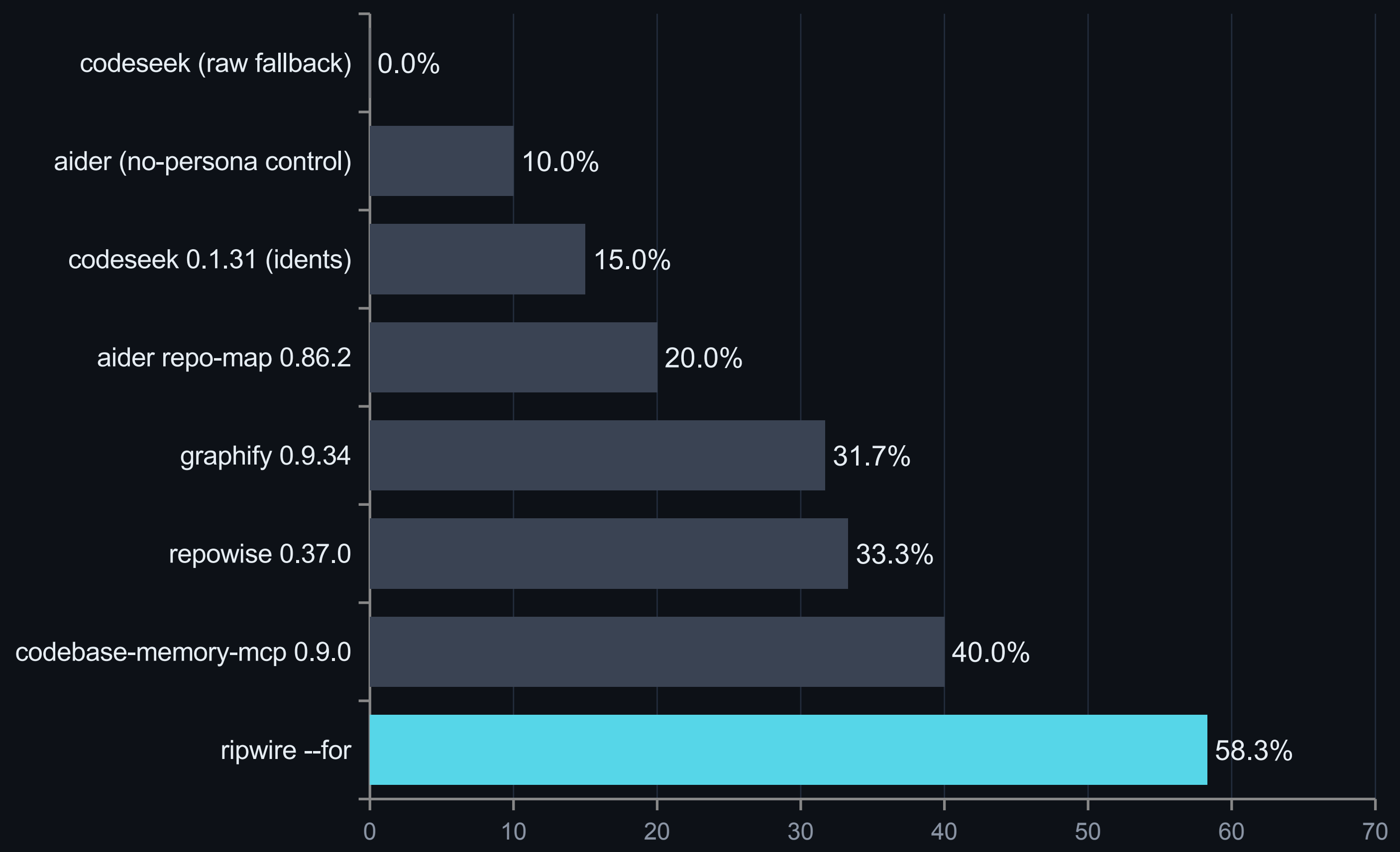
Every sweep row now carries a state (ok, rc=N, unparseable, unexpanded, timeout), a run where nothing answered refuses to report, and nothing it writes may land in the corpus. The cache sweep pins the root you are working in.

test/capsweepcheck.sh · 209f97a4 · 5723b2c0

every figure is quoted from its PR body or commit in the speaker notes — the tripwire covers our own instruments too

// one harness, one machine, one day — N = 60 paired, zero exclusions

Head-to-head: every competitor, one table



1.46×

the margin over the best competitor — 58.3% against 40.0% strict file@10

And the index it was measured with
ripwire 0.31 s · codebase-memory-mcp 1.24 s · codeseek 3.37 s · graphify 7.82 s · repowise 34.0 s, whose worst single index was **352 s**. Cold, nothing to an answer: 0.213 s.

What re-running cost us, published: aider moved 18.3% → 20.0% on its own tie-break nondeterminism, and we printed the higher number. Paired, ripwire loses 2 instances to codebase-memory-mcp, 2 to repowise, and 1 each to graphify, aider and the aider control.

bench/headtohead/r4-2026-08-06/ — harness, per-instance JSONL and recipe committed · replaces the two non-comparable tables r1 and r2 needed

// scored against a third-party oracle, never against each other

Graded by a compiler: zero silent misses

Both tools were scored against a scip-clang compiler-grade index of this repository — 68 answers, 34 blind-authored queries × --uses and --callers.

0

true silent misses — pre-fix and post-fix alike. Every imperfect answer either carried a discriminating self-flag (amb=, defs>1, external="1") or was an oracle-scope artifact where ripwire was right and the compiler could not see the file.

What a compiler-grade index costs

	ripwire	Serena 1.6.2.dev0 (clangd 19.1.2)	
warm query	49.8 ms	3,760 ms	~70×
cold start	208 ms	8.7 s	~40×
index build	none	~8 min	—
errors	0 / 101	7 / 61	—

Read honestly in both directions. The LSP is more precise — site-level 0.929/0.941 against ripwire's 0.914/0.941 after the fix round, a gap of roughly 1.5 points with recall now equal. ripwire also pays ~7× the bytes per call, because its output carries self-describing legends. What it cannot do is answer a macro query at all — clangd's documentSymbol has no macro entries, which is 3 of its 7 errors — or start in less than eight minutes.

The number that moved the wrong way, published with the rest: over-hedging rose 18.6% → 22.6% across the fix round. Closing loss buckets added self-flags faster than it removed imperfect answers. That direction is deliberate — calibrated to warn too often rather than too rarely — but it is a cost, and it is on the slide.

bench/headtohead/r9-2026-08-09/RESULTS.md — C++ only, 34 scorable queries, one corpus (this repository)

// 48 paired questions, zero exclusions, both arms exit 0 on all 48

Against a graph-database context server: 27–7–14

class	n	ripwire	GitNexus	tie	bytes ÷ it
symbol lookup	12	6	2	4	1.35×
conceptual search	15	8	1	6	1.23×
callers / blast radius	12	7	2	3	0.39×
task orientation	9	6	2	1	0.06×
all 48	48	27	7	14	0.159×

Its seven wins are named one by one, and its first-pass numbers do not exist. Four are cost losses on an answer ripwire gets right — a by-name lookup it serves in ~1 KB where ripwire spends 3× to also hand back the body. Three are ranking misses on webpack, and two of those share one unfixed mechanism: a symbol whose NAME matches the query beats the implementation that carries it in its body. Under the improve-first rule the round's FIRST pass was never published — its product was a loss list, and what is on this slide is the state after that list was worked. One of those fixes met its pre-registered band and was reverted anyway for failing a separate standing requirement; **the loss it would have fixed is still counted against us above.**

The cost of the whole sweep

309,348 B against 1,945,231 B. Warm median 197 ms against 1,082 ms. Index 0.25–0.45 s / 6.6–16.5 MB against 23–52 s / 391–623 MB — written into the repository, not $TMPDIR.

What it does better: compact one-hop context at 0.7–1.0 KB, a depth-LABELLED blast radius, per-stage timing on every query, and two-command onboarding into eight agents.

Failure modes, both directions

needed a 2nd call	0	8/48
empty radius, real callers	0	1
malformed through a pipe	0	7/48

The published numbers use the file-redirected form — the configuration favourable to it.

docs/EVALS.md §2 — GitNexus 1.6.9 (npm latest at 2026-08-22) vs ripwire at 7eb638e; django, webpack and this repository, all tree-hash verified clean · ratios are on class TOTALS, the medians form is printed there too because the two disagree

// the rule: numbers are held back until the loss list is worked

Every head-to-head round, and the ones that went against us

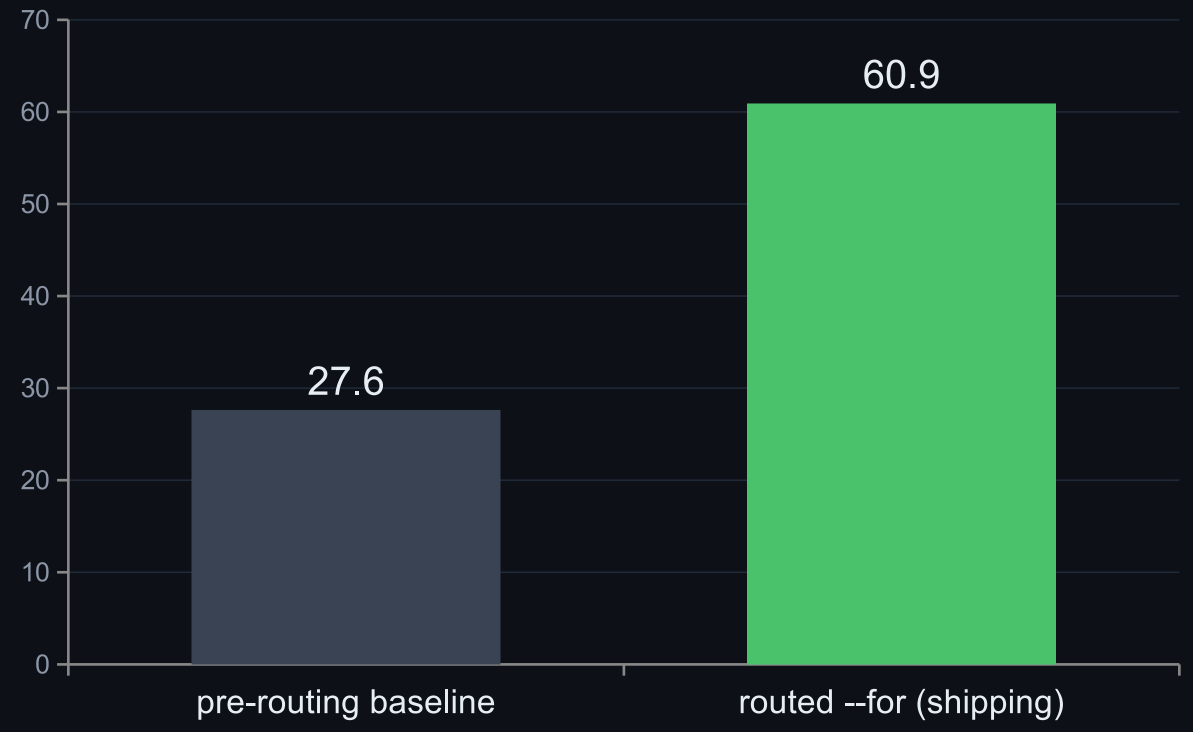
r1	2026-07-13/14	Aider repo-map · codebase-memory-mcp · graphify	*superseded by r4*
r2	2026-08-03	repowise · codeseek	*superseded by r4*
r3	2026-08-03	headroom — a compression layer, so a different instrument	passed code through byte-identical
r4	2026-08-06/08	all five competitors, one harness, one day	the table on the previous slide
r6	2026-08-04/05	agent-in-the-loop: the Codex CLI pilot	localization parity; +80% tokens, classified
r7	2026-08-08	CodeGraph, loss-first — we looked for our losses	fix REJECTED by its own preregistered band
r8	2026-08-08	aider again, at equal token budget	loss questions traced to one root cause
r9	2026-08-09	Serena / scip-clang oracle	0 silent misses; the fix list it produced
r10	2026-08-22	GitNexus 1.6.9 — a graph-database context server	27-7-14 — its own slide, two back

r7 in full, because it is the point: the fix was built, gated green, measured — predicted 12/22 against a pre-registered band of 10–14, measured 5/22 — and reverted rather than tuned. Published as a negative in docs/EVALS.md §7.

bench/headtohead/ (r1–r4, r9) · bench/r7/ · bench/r8/ — harnesses committed even for rounds that rejected their own fix · r5 left no head-to-head record in this tree, so it is not listed · r10's method and pins are published in docs/EVALS.md §2, its pre-fix numbers nowhere

// held-out, repo-disjoint, frozen dataset

Routing the ranker: +33 points, measured properly



+33.33pp paired delta, 243 held-out instances in 78 repository clusters

+25.00pp clustered-bootstrap 95% lower bound — the honest floor

−39.4% token ceiling for that gain

+3.4% warm latency for that gain

bench/locbench/ — czlll/Loc-Bench_V1, 560 rows, JSON SHA-256 pinned in dataset.lock; metric code = LocAgent's strict Acc@k

// compiled, not interpreted — and measured on public corpora

Fast enough to call on reflex

0.213 s

cold — nothing to an answer: parse, rank, reply, no cache (r4)

0.108 s

warm query, median — against aider's 2.920 s (r4)

0.31 s

index build — against repowise's 34.0 s, worst case 352 s (r4)

49.8 ms

warm query on this repo — against a clangd LSP's 3,760 ms (r9)

Where the second goes

Cold profiles are tree-sitter parse plus tag-query capture — not graph math. Resolve and PageRank together are a small fraction of the run, and the ranking was never the bottleneck.

Which is why the wins came from removing work, not micro-optimising it: demand-driven side captures, five whole-AST passes fused into one pre-order walk, a calibrated per-worker parse reserve.

Two opt-in faster builds

LTO is on by default. PGO is a script away and buys 14–25% cold.

Both are opt-in performance, not opt-in correctness: **the output stays byte-identical across build flavours** — the determinism gate runs on both, and CI builds a plain flavour too, because NDEBUG once compiled a whole class of degrade-path checks out of existence.

every card above comes from a committed harness — bench/headtohead/r4-2026-08-06/ and r9-2026-08-09/ — not from a private corpus

// how it got there — the pre-release performance sprint

Work removed before work optimized

file ingest

FILE* whole-file reads; CSV counting became a byte scan — iostream out of the hot path

capture gating

value-use capture runs only for the verbs that actually need it, not on every run

AST pass fusion

five whole-AST side-capture passes share ONE pre-order walk instead of five

parallel parse

query compile overlaps parse scheduling; files schedule by size — ids stay deterministic

radix + S+tree

numeric keys for edges and scores; a bounded-scratch hot map whose fixed cap reports its own overflow

no std::map

unordered_dense, reserve() before ingest — no std::unordered_map on any hot path

Why this slide carries no numbers. The sprint's own headline pair was measured in June 2026 on a large private C++ corpus that is not publicly reproducible, and the deck does not quote figures a reader cannot re-derive — the previous slide's numbers all come from committed harnesses. What survives the omission is the ***shape***: the graph math was never the bottleneck, and every win came from doing less work rather than doing the same work faster. Same map contract throughout — byte-identical output before and after each of these changes.

the mechanisms are in bench/PROFILE.md and the commit history; the private-corpus figures they produced are deliberately not quoted

// docs/EVALS.md §5, pinned on this repository 2026-08-08 (the --for row re-pinned 2026-08-22)

Ten everyday moments, and what each one costs

Ask it	Command	ripwire	naive read	savings
Orient me in this repo	`ripwire .`	~5.6K	~20K–25K	3.6×–4.5×
Where is X handled?	`--for="…"`	~2.1K	~4.9K–20K	2.3×–9.3×
What do I already know?	`--recall="…"`	~15K	~445K	29.2×
Set me up for this task	`--pack-task="…"`	~2.1K	~16K–80K	7.7×–37.7×
Show me this one function	`--expand=SYM --top-k=0`	~260–16.5K	~43K–174K	2.6×–670×
Who calls this function?	`--callers=SYM`	~580	~40K–52K	69.2×–89.1×
Is it safe to change this?	`--impact=SYM + --uses=SYM`	~1.3K	~18K	14.4×
I have a stack trace	`--from-trace=FILE`	~1.4K	~124K–298K	86.9×–208.6×
I changed these files — tests? radius?	`--situ`	~410	~3K–132K	7.3×–324.2×
Review this PR/diff	`--pr-context=REF`	~1.9K	~4.8K–51K	2.6×–27.5×

Figures are ~tokens (≈ bytes/4). Every row is scored same-correct-answer-or-it-doesn't-count — both sides were checked, not assumed — and every ratio is a **range**: the cheap end and the honest end of what an agent would actually read, never the single most flattering number. Absolute counts belong to the corpus AT THE PIN — .gitignore became a crawl default on 2026-09-03, which moved what a re-run reads; the commands, not the constants, are what reproduces.

// the cost lever

Smaller answers that are also better answers

74.7%

fewer element bytes at top-50 — the --pack-signatures ladder, root-neutralised, re-derived on this repo every run

Ask for a symbol by NAME and the router uses the name-exact ranker:

```
MRR  0.745 → 0.960
recall@1  61.1% → 91.3%
name-shaped queries on src/, all 3,011
doc-commented symbols scored, re-pinned
2026-09-05
```

This figure survived its own audit: three corrections deep, re-derived root-neutrally after the original was shown to depend on how the corpus path was spelled — three spellings of one root read 18.6 points apart before that subtraction. top-50 is the quotable number because the payload is top-50 whatever --top-k says.

And the pollution metric — fixture and generated paths contaminating results — is driven to 0.0% on the ranking lane while recall went UP, not down.

docs/EVALS.md §4–§5 — showcasecapturecheck re-derives the byte-reduction triple; `ripwire src --eval-retrieval` re-derives the ranking bars, scoring every doc-commented symbol rather than a sample

// the half no other tool ships

The tripwire: honesty as an output contract

counts_floor="1"

call edges come from source text — dynamic dispatch adds no edge, so every count that cannot claim completeness is labelled a floor, in the output itself

a zero is a measurement

zero means none found, never none exists — and absent ≠ 0

truncation is disclosed

every cap, page seam and budget cut is named in the header before you read a row

refusals teach

a refused query names the flag, the problem, and a working example — never a bare error

```
$ ripwire . --callers=rankGraphTeleport
<callers of="rankGraphTeleport"
  defs="1" count="6"
counts_floor="1">
<s t="fn" n="runEval" .../>
<s t="fn" n="rankGraph" .../>
…
</callers>
```

The map grades itself before it answers. This repository's own src/: files=153 symbols=5122 edges=14182 ambiguous=5982 unresolved=1598.

docs/EVALS.md §8 lists the numbers this project refuses to publish, each with its reason

// the honesty marks stop being a promise and become a measurement

Four levers accepted, one rejected on purpose

macro edges	kParserVer 48	function-like #define invocations land role="macro" edges on disclosed-degraded symbols
fn-pointer bindings	kParserVer 49	calls through unambiguous local function-pointer bindings now resolve
using-declarations	kParserVer 51	re-exports emit role="import" references — r9 loss bucket 1
shadow suppression	kParserVer 52	a local variable shadowing a function name stops stealing its use-sites — r9 loss bucket 2
RTA-lite	reverted	instantiation-filtered CHA cone: refuted TWICE — the instantiated set crossed namespaces, and narrowed calls LOST their amb= mark

The two r9 levers, against the compiler oracle: site precision 0.9046 → 0.9136, recall 0.9285 → 0.9412.

The fn-pointer lever raised unresolved= by 1,039. Disclosure, not regression: call sites the resolver now RECOGNIZES but refuses to resolve are counted instead of silently ignored.

bench/resolverround/RESULTS.md — per-lever attribution; RTA-lite's rejection is written up in docs/EVALS.md §7

// how it stays true

Proven, not promised

598 gate scripts

the suite runs on every push — plus determinism, cache-transparency and golden contracts; the gate count itself is gated against the runner's own loop

byte-identical, always

two runs over the same tree produce the same bytes; warm equals cold. Enforced in CI, twice — Release AND a plain flavour, because NDEBUG once blinded a whole class of checks

differential refactoring

a refactor must prove it changed nothing observable: two binaries, hundreds of argv vectors, stdout + stderr + exit codes byte-identical

held-out labels, authored blind

eval labels were written by reading source before the ranker ever ran on them — so the eval is allowed to say the ranker is wrong. It has.

sanitizer wall

ASan + UBSan + integer + float-cast, no-recover; TSan separately; leak suppression is one pinned file

its own output is audited

the showcase capture is regenerated and adversarially re-read as a claims corpus — the methodology ships in docs/METHODOLOGY.md

// the reason to believe any number on this deck

Claims you can trust, because we publish what failed

598

gate scripts named by test/regression.sh — and the COUNT itself is gated against the runner's own loop, so it cannot go stale quietly

8

registered NEGATIVES — changes built, gated green, measured against a band written before the code, and reverted rather than tuned

0

numbers on this deck without a committed instrument behind them — §8 lists the ones this project refuses to publish at all

`the fresh-clone arm`	our own battery only ever ran inside a configured, long-lived worktree — one value of every AMBIENT input. It clones HEAD into a genuinely unconfigured checkout and re-runs the gates whose past failures were exactly that: a developer's git config, a fixture's inherited branch name, the checkout's own path length.
`the merge-conservation gate`	consistency is not conservation. A conflict resolution can drop a gate from the runner's loop while regenerating the pinned count from that SAME damaged list — both sides then agree at a wrong number. On a merge, the loop must equal the UNION of both parents', minus explicit tombstones.
`the eight, by name`	r7's prose-strip fix (predicted 12/22, band 10–14, measured 5/22) · RTA-lite, refuted twice · file-level evidence pooling, +0.00pp held-out · un-guarded query stemming · its IDF-guarded retry, VOIDED by its own verifier · query-term density weighting, both displacement guards tripped · --anchor and --cochange-boost, no confirmed lift · a name-coverage floor that MET its pre-registered band and was reverted anyway for a standing requirement.

A gate that cannot go red is a gate that proves nothing — which is why every arm on this deck ships with the mutation that makes it fail.

// own the losses — they are in the README too

Where it loses, published with the wins

The public C++ number is lower than the private one it replaced	SFML: strict file@10 28.7%, any@10 41.7%, first-hit MRR 0.21. Until 2026-08-07 this bullet compared against a ~89% any@10 private corpus that is no longer reproducible. That comparison is retired; the public number is the baseline going forward.
--grep costs more than it saves	+19.7% tokens / −11.2% — published next to the flag it indicts, and carrying the same private-corpus caveat as every figure in that source
PageRank is the wrong co-change ranker	3.8% recall@5 against 40.3% for plain lexical, and fusing the two made it worse — relatedness is lexical, importance is structural
Strict multi-file localization stays hard	single-file gold 73.4% vs multi-file 18.2% (held-out); the same cliff on every corpus — open headroom, said plainly

A tool that publishes its counterexamples is a tool whose wins you can believe.

// the single command for "is this code actually rotten?"

Six families of evidence, counted — never blended

structural	shape: complexity, size, nesting, params, readability rank	--metrics
lexical	identifier text: the naming rules	--lint --naming-consistency
confusion	syntax: the atoms-of-confusion rules	--lint
historical	git change frequency, measured PER FILE	--hotspots
colocation	how much you must READ outside this file	--context-ratio
state	this function's OWN BODY touching non-local mutable state	--nonlocal-state

Ranked by the COUNT of distinct families that fire — never a weighted composite, because a composite lets one loud family impersonate six.

4 of 6 is what the strict preset counts — the verb says so itself, families="6" enabled_n="4". Two families did not clear the stability ladder.

`--quality-panel[=strict|default|lenient]` — and `--quality-delta` for the narrower question: what did MY change make worse?

// a new lens does not ship because it sounds plausible

The calibration that disqualified a family

+0.168

the LARGEST cross-family correlation in the whole 6×6 matrix — widening the panel from four families to six did not raise it

27,889

eligible functions, five independent trees — two reported separately and never pooled, because one of them is this repository

4 of 6

families the strict preset actually counts — TWO were measured and held back from gating

The uncomfortable result, kept. The stability pass ran each family across a ladder of commits to ask whether it is steady enough to gate on. **colocation came out worse than historical on one ladder**, so neither gates: strict counts four of the six — a lens its own author wanted, measured, and demoted. The ladder was run precisely because that answer was possible; assuming it would pass is how a plausible axis becomes a permanent one.

--field-affinity was NOT made a family either — an exclusion by unit, not a failed measurement: it ranks struct FIELDS by co-access against declared layout, and the panel's unit is a function. Stated in the calibration rather than left for someone to notice.

docs/EVALS.md §9.9 — same harness and criteria as the original four families, run BEFORE the two new ones shipped enabled

// examples from docs/QUALITY_DELTA_CATALOG.md — each one reproduced, none invented

PLACEHOLDER

What --quality-delta catches

‹kind› ‹the finding, as the tool prints it›

before	after
‹before›	‹after›

‹why it matters, in one sentence›

‹kind› ‹the finding, as the tool prints it›

before	after
‹before›	‹after›

‹why it matters, in one sentence›

‹kind› ‹the finding, as the tool prints it›

before	after
‹before›	‹after›

‹why it matters, in one sentence›

placeholder: the examples arrive as data (QD_EXAMPLES, top of the generator) once docs/QUALITY_DELTA_CATALOG.md lands

// built for the agent's seat

One command wires it into your agent

```
$ ripwire wrap claude
```

claude · cursor · codex · windsurf · gemini · aider — or --all to detect every one you have installed

31 MCP verbs

16 read verbs mirroring the CLI, 12 flagship reflexes (impact, uses, edit_check, from_trace, connect …), 3 span-addressed edit verbs with a safety contract

lazy-body handles

read verbs return signatures and a stable handle; the agent fetches a body only when it decides it needs one — names by default, bytes on request

18 agent skills

moment-matched workflows (orient, navigate, change-check, quality-bar …) — wrap prints the recipe, skills/install.sh installs them

11 orchestrator loops

copy-paste prompts in prompts/: run the same audit, eval and head-to-head machinery that built this tool, on your own repository

wrap does not just print JSON: it probes what that agent already has, emits the config in that agent's own shape, and includes a use-when blurb so the agent knows which verb fires at which moment — not just that a server exists.

the MCP server exposes the same deterministic engine — one index, shared with the CLI, staleness-checked

// it does not only read

Span-addressed edits: the map writes back

Three verbs, one resolved span

```
--replace-symbol-body=TARGET
--insert-before-symbol=TARGET
--insert-after-symbol=TARGET
  --edit-payload=FILE|-
```

TARGET is a name, an @FILE:LINE seed — paste the location out of a diff hunk or a compiler error and it binds the innermost enclosing definition — or a freshness-pinned handle from --grep --handles. No line numbers to keep in your head, and no whole-file rewrite.

The refusals are the feature

Freshness hash, lock, a re-check immediately before the rename, fsync, mode preserved, atomic rename. A target that resolves to more than one definition refuses. An empty payload refuses rather than being read as a deletion. Every refusal leaves the file byte-identical, and every success prints a JSON receipt whose span is the POST-EDIT byte range — where the payload now sits, not where it was aimed.

Preview the bytes before they exist

--edit-check=SYM --edit-payload=- --dry-run splices the payload over the span IN MEMORY, re-parses that one file, rebuilds the call graph over the re-derived tree, and answers the contract question about bytes nobody has written: did the signature change, and which callers does it break? Preview, then apply — no Read of the file in between.

Several edits, one transaction

--edit-plan=FILE is a versioned JSON multi-edit plan and its mode is EXPLICIT: exactly one of --dry-run or --apply, and neither is not a default. Every target, payload and span is preflighted before any write; overlapping edits refuse; payload paths are confined to the plan's own directory, so an absolute path or a '..' escape refuses and names what it resolved to.

Same engine on both seats: the three MCP edit verbs are these three flags, and the navigation the deck has no room for is the same story — --graph-query, --whereis, --stray-content, --at, --arch, --owners, --export, --help-task all sit in docs/COMMANDS.md with a recorded invocation and its output.

every claim on this slide is stated by `ripwire --help` and captured in docs/COMMANDS.md — test/deckcheck.sh proves each flag named here exists in the shipped binary

// standing on giants

The research inside — classic and current

49 repositories + 70 papers folded · and a labelled survey of 237 tools that contributed nothing, which says so — every row with the lesson taken and where it lives: docs/LINEAGE.md

1976	**Cyclomatic complexity — McCabe** the metric behind --hotspots
1994	**Okapi BM25 — Robertson · Spärck Jones** the lexical ranker (twice: subtoken+body, whole-name)
1998	**Personalized PageRank — Page · Brin** the headline importance signal
2008	**Louvain modularity — Blondel et al.** the module / community map
2009	**Reciprocal Rank Fusion — Cormack et al.** deterministic signal fusion (--rank-by=rrf)
2011	**Readability model — Posnett · Hindle · Devanbu** the lexical family of the quality panel
2019	**Delta Maintainability Model — di Biase et al.** --dmm: one comparable number per change

TDAD · arXiv 2603.17973

a static test-map cut agent-caused regressions 6.08% → 1.82%; prose TDD instructions alone made agents WORSE → the shape of --test-gate

What-to-Retrieve · arXiv 2503.20589

retrieval selection beats retrieval volume for coding agents → the selection-over-dumping thesis

LocAgent / Loc-Bench (2025)

the localization metric (strict Acc@k) and the frozen 560-instance dataset every accuracy number here is scored on

scip-clang · Serena / clangd

the compiler-grade oracle round 9 graded both tools against — an outside referee, not a rival

tree-sitter

the incremental GLR parsing substrate — 23 grammars vendored, one parser per thread

counts gated in-repo: readmedriftcheck arm E derives them from LINEAGE.md's own tables — the lineage is the design rationale

// do not take any of it on trust

Every claim, and the command that re-derives it

179 long flags · 33 slides	`bash test/deckclaimcheck.sh`
every --flag named here exists	`bash test/deckcheck.sh`
74.7% fewer element bytes	`bash test/showcasecapturecheck.sh`
598 gate scripts	`bash test/manifestcheck.sh`
49 repos · 70 papers · 237 surveyed	`bash test/readmedriftcheck.sh`
the ten moments, any row	`ripwire . --callers=SYM | wc -c`
the head-to-head table	`bench/headtohead/r4-2026-08-06/`
the oracle round	`bench/headtohead/r9-2026-08-09/RESULTS.md`

The deck is generated, and the generator is gated. present/deck5_ripwire_build.js is scanned for fabricated flags and its slide count is derived from its own addSlide() calls — a deck that drifts from the binary fails the suite.

docs/EVALS.md is the source of truth; §7 is the counterexamples, §8 the claims this project refuses to make

// sixty seconds to the first ranked map

Quickstart

```
git clone <repo> && cd ripwire
cmake -S . -B build && cmake --build build -j      # hermetic: works with the network off
./build/ripwire --help
```

`ripwire .`	the ranked map — start here on any unfamiliar repo
`ripwire . --for="cache invalidation"`	the task lens: what to touch, ranked
`ripwire . --callers=someFunction`	who calls it — with the honesty marker attached
`ripwire . --quality-panel`	is this code actually rotten? six families, one shortlist
`ripwire . --test-gate`	before you commit: exactly which tests must run
`ripwire wrap claude`	wire it into your agent, in that agent's own config shape

ripgrep for the speed. **tripwire** for the honesty. Apache-2.0.